

12.421/12.621 Lecture 13

Radiometry and radiative transfer

Topics covered

- Review of black body radiation
- Overview of radiometry
- Radiative transfer equation

Radiometry

- Remote sensing method for measuring EM radiation from planetary surfaces, atmospheres, and other natural media

→ More specifically, radiometry is measure of incoherent radiant EM energy

- To get started, recall Planck's Law from last lecture

$$B_\omega = \frac{2hf^3}{c^2} \left[\frac{1}{e^{\frac{hf}{k_B T}} - 1} \right] \quad \left[\frac{W}{m^2 sr Hz} \right] \equiv \text{spectral radiance}$$

$$h \equiv \text{Planck's const} = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$f = \frac{\omega}{2\pi}$$

$$c \equiv \text{speed of light}$$

$$k_B \equiv \text{Boltzmann's const} = 1.38 \times 10^{-23} \frac{J}{K}$$

$$T \equiv \text{absolute temperature}$$

- Recall that when we increase T , we increase the frequency of max B_ω (equival, we decrease the wavelength according to $\lambda_{max} \propto T^{-1} \rightarrow$ Wien's displacement Law) and total radiated power (area under B_ω curve)

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• How is this used in practice?

↳ First note that B_w is the power emitted by a black body with a 1 m^2 surface area through a 1 sr solid angle over a bandwidth of 1 Hz

↳ Suppose black body has area A_s , the spectral power P_w received by an antenna w/ area A_r at a distance R from black-body is

$$P_w = B_w A_s \Omega_r \left[\frac{\text{W}}{\text{Hz}} \right]$$

where $\Omega_r = \frac{A_r}{R^2}$ is solid angle subtended by the antenna

↳ Now note that source area subtends a

$$\text{solid angle } \Omega_s = \frac{A_s}{R^2}$$

$$\Rightarrow R^2 = \frac{A_r}{\Omega_r} = \frac{A_s}{\Omega_s}$$

and

$$P_w = B_w A_r \Omega_s$$

↳ Total power received by a receiver w/ bandwidth

$$\Delta f = f_{\text{high}} - f_{\text{low}}$$

$$P = \int_{f_{\text{low}}}^{f_{\text{high}}} P_w df = A_r \Omega_s \int_{f_{\text{low}}}^{f_{\text{high}}} B_w df$$

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↳ For sufficiently large Δf and an appropriate center frequency, $P \propto$ total intensity of black body given by Stephan-Boltzmann Law (see Lecture 12)

$$I \propto \pi P = \sigma T^4$$

where

$$\begin{aligned} \sigma &\equiv \text{Stefan-Boltzmann const} \\ &= 5.67 \times 10^{-8} \left[\frac{\text{W}}{\text{m}^2 \text{K}^4} \right] \end{aligned}$$

* But be careful, some RS instruments have narrow bandwidth $\Rightarrow$ Stefan-Boltzmann may not apply

- For relatively narrow bands of most RS instruments, must consider Planck's Law or an approximation of it
 - $\Rightarrow$ at infrared frequencies, it may be advisable to use Planck's Law
 - $\Rightarrow$ at microwave frequencies, can use Rayleigh-Jeans Law

$$B_\omega = \frac{2k_B T f^2}{c^2}$$

$$\left[\begin{array}{l} \text{To get Rayleigh-Jeans from Planck's consider identity} \\ e^{\delta} - 1 = \left(1 + \delta + \frac{\delta^2}{2!} \right) - 1 \quad \left(\text{where } \delta = \frac{hf}{k_B T} \right) \\ \approx \delta \quad \text{for } \delta \ll 1 \rightarrow \text{low frequency limit} \\ \text{for which Rayleigh-Jeans applies} \end{array} \right]$$

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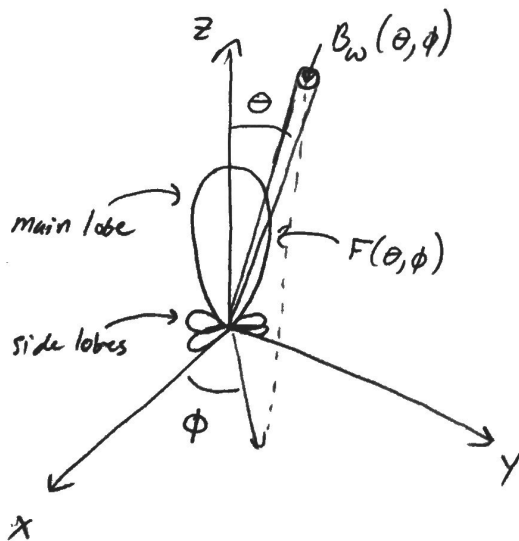
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- To understand relationship between power received by radiometer and the temperature of source body, we must account for directionality of antenna that has a radiation pattern represented by an arbitrary function $F(\theta, \phi)$ where θ is the colatitude and ϕ the longitude (spherical coordinates)

$F(\theta, \phi)$ represents the antenna gain as a function of direction

→ In the case of radiometry, $F(\theta, \phi)$ indicates the antenna's ability to convert EM radiation coming from direction defined by θ and ϕ into electrical power (signal)

- Most RS antenna will have a lobe structure to the radiation pattern $\Rightarrow$ observational geometry



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- To account for antenna directivity let's consider the differential spectral power

$$dP_\omega = B_\omega A_r F(\theta, \phi) d\Omega$$

Then the total power received by antenna over bandwidth $\Delta f = f_2 - f_1$

$$P = c_p A_r \int_{f_1}^{f_2} \iint_{4\pi} B_\omega F(\theta, \phi) d\Omega df$$

where c_p accounts for polarization of antenna

$$c_p = \begin{cases} 1 & \text{non-polarized antenna} \\ \frac{1}{2} & \text{polarized antenna} \end{cases}$$

Why does $c_p = \frac{1}{2}$ for polarized antenna?

Because EM wave can be represented by two orthogonal polarizations, each w/ half of the total power. (Think: polarized sunglasses) Black bodies radiate unpolarized EM energy $\Rightarrow$ polarizing filter cuts total power in half

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- Now assuming Rayleigh-Jeans law is a valid approximation (which is true for microwaves), total power received from a black-body

$$P_{bb} = c_p A_r \int_{f_1}^{f_2} \iint_{4\pi} \frac{2K_B T}{\lambda^2} F(\theta, \phi) d\Omega df$$

- Taking the common case in RS of a narrow bandwidth where $B_\omega \approx \text{constant over } \Delta f$

$$P_{bb} = K_B T \Delta f \underbrace{\frac{A_r}{\lambda^2} \iint_{4\pi} F(\theta, \phi) d\Omega}_{\Omega_p} \quad \text{for } c_p = \frac{1}{2}$$

$\Omega_p \equiv \text{antenna pattern solid angle}$

- From antenna theory (Ulaby and Long, 2014, ch 3)

$$\Omega_p = \frac{\lambda^2}{A_r}$$

$$\Rightarrow \boxed{P_{bb} = K_B T \Delta f} \rightarrow \text{power received at antenna}$$

is proportional to T w/ const of proportionality

$= K_B \Delta f$ (one a universal const the other an instrument parameter, both known)

$\Rightarrow$ common in radiometry to interchange terms 'power' and 'temperature'

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- Now, a black-body is an idealized material;
Real materials emit less radiation than black bodies (called gray bodies; white bodies reflect all incident radiation)
- And in RS applications, it is often the case that observed EM radiation has propagated through some medium (e.g., we may be interested in the T of a fire as seen from a satellite) that absorbs and/or scatters EM radiation
- Let's take each of these factors in kind
- Let us consider a semi-infinite material w/ brightness (intensity) $I(\theta, \phi)$ and physical (or actual) temperature T . The observed brightness

$$I(\theta, \phi) = e(\theta, \phi) \frac{P_{bb}}{\lambda^2}$$
$$= \frac{k_B}{\lambda^2} \Delta f e(\theta, \phi) T = \frac{k_B}{\lambda^2} \Delta f T_B(\theta, \phi)$$

where

$T_B = eT \equiv$ brightness temperature

$e(\theta, \phi) \equiv$ emissivity, defines material's ability to emit EM radiation

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- In terms of power (per unit area), emissivity is

$$e(\theta, \phi) = \frac{I(\theta, \phi)}{I_{bb}} = \frac{P(\theta, \phi)}{P_{bb}}$$

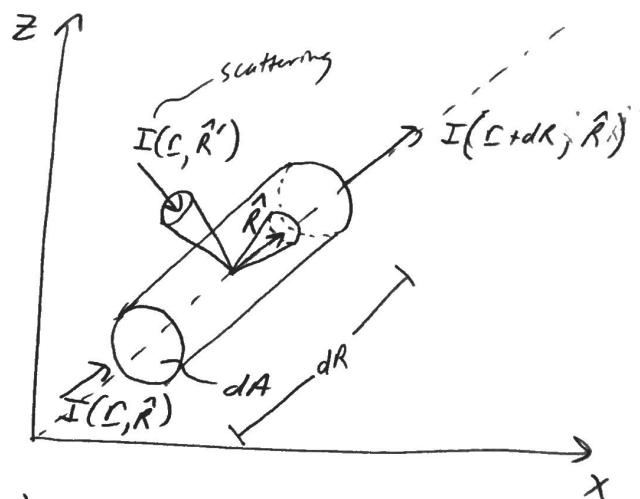
$$\Rightarrow 0 \leq e(\theta, \phi) \leq 1$$

↑ white body ↑ black-body (total absorption and emission)
(nothing absorbed $\Rightarrow$ nothing emitted)

- Brightness temperature often used as a proxy for observed brightness and (equiv.) power because of the proportional relation w/ a known constant of proportionality
- Let us relate T_B to the physical and dielectric properties of the material and atmosphere
- As we have been discussing, interaction of EM radiation w/ matter involves emission (source) and absorption + scattering (extinction; sink)
- Radiative transfer deals w/ these different interactions

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- To derive a simple function for radiative transfer, let us consider EM energy propagating in direction $\hat{R}$ through an infinitesimal cylinder w/ cross-sectional area dA , length dR and position given by vector $\underline{r}$



$I(\underline{r}, \hat{R}) \equiv$ incident brightness at position $\underline{r}$
in direction $\hat{R}$ $\left[\frac{W}{m^2}\right]$

$I(\underline{r}+d\underline{r}, \hat{R}) \equiv$ brightness of EM radiation leaving
cylinder $\left[\frac{W}{m^2}\right]$

$I(\underline{r}, \hat{R}') \equiv$ brightness of EM radiation scattered
into the cylinder from surrounding area

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- Power (brightness) leaving cylinder is $<$ that entering cylinder due to extinction (scattering + absorption)
- The change from incident brightness $I(\underline{r}, \hat{r})$ due to extinction is

$$dI_{\text{extinction}} = K_e I(\underline{r}, \hat{r}) dR$$

where

$K_e \equiv$ extinction coef (introduced in Lecture 9)

$$= K_a + K_s$$

$$\begin{aligned} &\quad \quad \quad \hookrightarrow \text{scattering coef} \\ &\quad \quad \quad \hookrightarrow \text{absorption coef} = -2K_0 \text{Im}\{\sqrt{\epsilon}\} \end{aligned}$$

- Media inside cylinder emits EM radiation w/ brightness

$$dI_{\text{emission}} = K_e J(\underline{r}, \hat{r}) dR$$

where

$J \equiv$ total source function

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- Total source function J is a weighted sum of scattering J_s and absorptions (emission) J_a source functions

$$J = (1-a) J_a + a J_s$$

where $a = \frac{\chi_s}{\chi_e} \equiv \text{single-scatter albedo}$

$J_s \equiv$ scattering source function: accounts for radiation scattered along $\hat{R}$ (by matter in cylinder) and sourced from EM radiation coming in from all directions (e.s., sunlight scattered by surrounding atmosphere)

$J_a \equiv$ absorption source function: emission must balance absorption when matter is in thermal equilibrium (Kirchhoff's Law of thermal radiation) $\rightarrow$ In RS, timescales are sufficiently short that this is a reasonable approx.

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- Now, propagation over distance dR results in change of (power) intensity dI

$$dI = \overset{\substack{\text{source} \\ \downarrow}}{dI_{\text{emission}}} - \overset{\substack{\text{sink} \\ \downarrow}}{dI_{\text{extinction}}}$$

$$= \kappa_e J dR - \kappa_e I dR$$

$$= \kappa_e dR [J(\underline{r}, \hat{R}) - I(\underline{r}, \hat{R})]$$

- Defining optical depth $d\tau = \kappa_e dR$, we can rearrange to get the radiative transfer function

$$\boxed{\frac{dI}{d\tau} + I = J}$$

In words: Change in brightness w/ optical depth (product of κ_e and dR) = total source less incident brightness temperature

↳ In essence, statement of conservation of energy

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- Relating to brightness temperature T_B

Recall Rayleigh-Jeans

$$I = \frac{K_B}{\lambda^2} \Delta f T_B$$

By Kirchhoff's Law, emission = absorption:

$$\Rightarrow J_a = \frac{K_B}{\lambda^2} \Delta f T \quad T = T(\underline{r})$$

$$\hookrightarrow J_a \text{ is isotropic} \Rightarrow J_a = J_a(\underline{r})$$

Analogously, define volume scattered radiometric temperature $T_{Vs}(\underline{r}, \hat{r})$ such that

$$J_s(\underline{r}, \hat{r}) = \frac{K_B}{\lambda^2} \Delta f T_{Vs}(\underline{r}, \hat{r})$$

Plugging into radiative transfer function

$$\frac{dT_B}{dt} = (1-a)T + aT_{Vs} - T_B$$